

Supplemental Material for

A Controlled Study of HyDE-Style Query Expansion for Multi-Hop Evidence Acquisition

This supplement is organized as an auditable companion to the main manuscript. Tables S1–S7 provide the core lookup tables cited from the main text: configuration provenance, dataset artifact provenance, HotpotQA and 2Wiki paired statistics, deduplication sensitivity, guarded canonicalization decomposition, and full call/prompt-character accounting. Table S8 lists baseline implementation details; Tables S9–S10 report model/API provenance; and Table S11 gives the full corpus-scale retrieval stress test. Tables S12–S13 define answer and evidence metric normalization before the answer-overlap diagnostics. Tables S14–S20 collect answer-overlap, gold-content, and 2Wiki query-composition diagnostics. Table S21 reports query-length matched sensitivity, and Table S22 gives the single-annotator error taxonomy. Tables S23–S28 cover corpus statistics, supporting-title completeness, artifact availability, qualitative cases, and the supporting-paragraph audit. Table and section counters are independent: references always specify either *Table Sx* or *Section Sx*. Unless otherwise stated, artifact paths are relative to the anonymized project root.

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Table S1: Configuration provenance for the reported processed-data evaluation.

Component	Selection basis	Data inspected	Frozen stage
$q + h$ HyDE query composition	Chosen from the HyDE-style design rationale: keep the original question as an anchor and append a document-like hypothetical passage for dense retrieval. It is not claimed to be selected by optimizing over the reported q -only, h -only, or $q + h$ rewrite diagnostic rows.	Released processed-split schema and local retrieval artifacts were inspected to implement the pipeline. The reported q -only, h -only, and $q + h$ rewrite diagnostic results were generated after the final $q + h$ protocol was already in use.	Canonical $q + h$ HotpotQA retrieval and reader artifacts were frozen on 2026-07-04. Query-composition controls were added later: q -only and h -only on 2026-07-06, and $q + h$ rewrite on 2026-07-08.
HyDE prompt	Template asks for one concise hypothetical supporting passage for retrieval-query expansion. The design follows document-like HyDE-style expansion rather than a prompt search over final EM/F1.	Prompt-template design, released question text, and later generated-passage artifacts for answer-string-overlap auditing. Gold answers are retained for evaluation metadata but are not inserted into the prompt text.	Frozen before canonical HotpotQA HyDE generation. Prompt-builder and prompt-file hashes are reported in Table S10.
Rewrite prompt	Control prompt designed to spend one extra query-side Qwen3.7-Max call while producing a direct retrieval query rather than a document-like passage.	Same reported question set and control-design constraints. Gold answers are not inserted into prompt text. The prompt is a matched-call control, not an optimized alternative final method.	Frozen before query-reformulation generation and reader scoring on 2026-07-08 to 2026-07-09; prompt-builder hash is reported in Table S10.
Equal-budget query-form prompts	Post-hoc mechanism diagnostic added after the primary HotpotQA analysis to test whether document-like form remains useful at similar generated length and call budget. The four modes use a shared one-call, 40-word target, 35–45 word instruction window, and 64-token generation cap.	Primary HotpotQA results and the length-versus-form threat were inspected before designing this diagnostic. After the prompt family was fixed, the same templates and budgets were applied to HotpotQA and 2WikiMultihopQA without dataset-specific modification. Gold answers are retained as scoring metadata but are not inserted into prompt text.	Prompt JSONL files were generated on 2026-07-21. Full equal-budget Qwen3.7-Max generations were frozen on 2026-07-22 to 2026-07-23, followed by local BGE-base retrieval scoring on 2026-07-23. This row is a post-hoc robustness diagnostic, not part of the primary frozen study.
Reader prompt	Generic evidence-grounded short-answer prompt with common answer-type instructions for yes/no, location, date/age comparison, role/title, and answer-only formatting.	General multi-hop QA answer-format requirements and retrieved evidence serialization. The same reader prompt is used across retrieval variants and reused on 2WikiMultihopQA without dataset-specific retuning. The reader is not constrained to copy a contiguous evidence span.	Frozen before final reader prompt JSONL and answer artifacts were generated; prompt-builder hash is reported in Table S10.
Top- $k = 10$ retrieval budget	Common evidence budget inherited from the processed-data retrieval-reader protocol and used for comparable retrieval metrics and reader prompts.	Released IRCot processed-data format, local candidate-pool size, and prompt-length feasibility.	Applied uniformly to all reported methods and both datasets before final scoring; no method-specific top- k tuning is reported.
900-character evidence truncation	Prompt-budget rule for comparable reader and verifier evidence blocks under hosted-model context limits.	Local corpus paragraph lengths and prompt serialization constraints were inspected during prompt construction.	Applied before final reader and verifier prompt JSONL artifacts were generated; the same limit is used for reader and verifier evidence blocks.
Verifier risk rules	Secondary verifier selection policy for likely format, alias, date, count, granularity, abstention, role/type, comparison, overlong-answer, and short-entity risks.	Same HotpotQA split diagnostics over canonical HyDE reader outputs and EM mismatches. This is disclosed as same-split verifier development, not as independent held-out tuning.	Frozen before final HotpotQA HyDE verifier prompt generation; applying the frozen policy selected 120 verifier cases.
Verifier prompt	Conservative JSON verifier prompt with <i>keep</i> as default and a limited set of benchmark-style corrections.	Same HotpotQA verifier-development diagnostics and retrieved-evidence examples used to define safe correction categories.	Frozen before the 120 verifier calls; all 120 structured outputs are audited in Table S10.
Abstention guard	Deterministic merge rule to block a new abstention when the initial reader answer is not an abstention.	Same HotpotQA verifier diagnostics and prior harmful-abstention risk analysis.	Frozen before the final guarded merge. It is inactive on the final HyDE verifier outputs but retained as a preventative guard.
Numeric-unit guard	Deterministic merge rule to block unsupported additions of units to bare numeric answers.	Same HotpotQA verifier diagnostics and raw-verifier transition analysis, including observed correct-to-wrong numeric-unit expansions.	Frozen before the final guarded merge; the guard removes the observed correct-to-wrong transitions in the reported HyDE verifier reconstruction.
BGE-base stronger-encoder reproduction	Post-hoc robustness check asking whether the query-composition pattern persists with a stronger dense encoder. It is not used as a tuned replacement for the MiniLM primary protocol.	Primary MiniLM results and frozen generated query artifacts were inspected. The diagnostic changes the encoder to BAAAT/bge-base-en-v1.5 while keeping prompts, top- k , reader prompt, and local-corpus protocol fixed.	Added after the primary HotpotQA scoring pass; reported as a robustness reproduction rather than a pre-specified primary configuration.
Gold-aware scrubbing and answer-overlap diagnostics	Post-hoc internal-validity diagnostics for query-side answer-string overlap and generated gold-evidence traces.	Generated hypothetical passages, gold answers, supporting titles, and frozen retrieval artifacts were inspected for diagnostic grouping or scrubbing. These analyses are not used to choose a new deployed prompt.	Added after the primary HotpotQA results to bound the answer-leakage explanation; reported as diagnostic evidence only.
Corpus-scale and hard-negative audits	Post-hoc retrieval-only stress tests for corpus size, added-gold-title leakage, near-duplicate overlap, and harder processed-train distractors.	Released evaluation corpus records, processed-train distractors, frozen query-generation artifacts, and BGE similarity scores were inspected. The hard-negative row samples from the audited processed-train 100k pool rather than full Wikipedia.	Corpus-scale stress artifacts were generated after the primary study; leakage-filtered and BGE-hard audit artifacts are retained under the corpus-scale audit directory. These rows test external-validity threats after the primary study.
Qwen-Turbo fixed-evidence reader check	Post-hoc targeted reader robustness check. It changes only the reader model over fixed equal-budget top-10 evidence rather than rerunning query generation or retrieval.	Equal-budget retrieval outputs for direct rewrite, keyword/entity expansion, and document-like passage were inspected; no new retrieval artifacts were selected based on Qwen-Turbo answer quality.	Qwen-Turbo answer artifacts were frozen on 2026-07-23 to 2026-07-25. The check is a limited second-reader readout, not a broad reader sweep or primary end-to-end claim.

This table separates selection basis, inspected data, and freeze stage. It does not claim that every configuration was selected on an independent development split. The primary MiniLM retrieval-reader comparison is artifact-frozen as the main study; BGE, equal-budget, scrubbing, corpus-scale, top- k , supporting-paragraph, and Qwen-Turbo rows are post-hoc robustness diagnostics; same-split tuning is explicitly limited to the secondary verifier selection and guarded-merge analysis.

S1 Configuration Provenance and Freezing Protocol

The manuscript uses released IRCot processed HotpotQA and 2WikiMultihopQA `test_subsampled` splits under a processed local-corpus protocol. The following records separate retrieval-reader protocol choices from verifier-specific diagnostic choices by listing the selection source, the data inspected, and the artifact-level freeze evidence for each configuration. They do not claim that every design choice was selected using an independent development split. In particular, the $q + h$ row supports a narrower provenance claim: the final HyDE query composition was fixed before the reported q -only, h -only, and question-plus-rewrite diagnostic controls were generated and scored. Same-split diagnostic tuning is disclosed for the secondary verifier risk rules and guarded-merge policy.

Table S2: Dataset source and converted-artifact provenance for the processed local-corpus protocol.

Dataset	Upstream repository and commit	Source file and local download	Raw SHA-256	Converted JSONL SHA-256
HotpotQA	github.com/ StonyBrookNLP/ircot commit: 3c1820f698eea5eed db4fba3c56b64c96 1e063e4 main; 2024-06-11	processed_data/ hotpotqa/ test_subsampled.jsonl downloaded: 2026-06-30 21:01:24	729D8987560092C8 476523488069E6FB 9BD1527157DC0D7B FEBF3DDC336C558E	23F4AD6F82182C1B FE34C7F42DA3E852 questions: 125EA71B0B9138CC corpus: B3E1740B3501E8AA 8B27434696F9AB47 9EB7301B7CB51259 FDFBD9C7CEA96D7A 2A728B68F761ACB2
2Wiki	github.com/ StonyBrookNLP/ircot commit: 3c1820f698eea5eed db4fba3c56b64c96 1e063e4 main; 2024-06-11	processed_data/ 2wikimultihopqa/ test_subsampled.jsonl downloaded: 2026-06-30 21:01:21	4948757B90627F40 B77CD6BF81A62C1A 7AD525D6A6A0C91B 17A2E0E2FA88F5BC	7E1871FE8813606A questions: 1E4FBE393A8A1C48 corpus: CC67376CBB2E864F 4F8926A37A3FF1C7 CF7603B72D3B0BA0 7F008C6FEEB281E 2C63579098CF98F2 C1E47E61F241069C0

Converted artifacts are questions.jsonl and corpus.jsonl under local-artifacts/datasets/ircot_hotpotqa_test500 and local-artifacts/datasets/ircot_2wikimultihopqa_test500. HotpotQA converted files have local timestamp 2026-06-30 21:17:21; 2WikiMultihopQA converted files have local timestamp 2026-07-04 22:15:50.

The conversion script is src/data/prepare_ircot_hotpotqa.py, SHA-256 prefix C23D8CC9E6EB. It reads each processed row's contexts, writes one question record with the first extracted answer and supporting-title list, and writes one paragraph-level corpus record per context with doc_id, title, paragraph text, source question id, and is_supporting. The local corpus is then formed as the within-dataset union of these question-level candidate pools.

Table S3: Full HotpotQA paired statistics on the released IRCot test_subsampled split.

Endpoint	Baseline $\rightarrow$ target	Δ metric 1 [95% CI]	Δ metric 2 [95% CI]	$W \rightarrow C / C \rightarrow W$	Test p
Answer EM/F1	One-step Dense RAG $\rightarrow$ HyDE-style RAG	0.0900 [0.0620, 0.1200]	0.0906 [0.0634, 0.1192]	53 / 8	< 0.0001
Answer EM/F1	Single-query Reformulation RAG $\rightarrow$ HyDE-style RAG	0.0880 [0.0600, 0.1180]	0.0976 [0.0690, 0.1266]	52 / 8	< 0.0001
Answer EM/F1	Question + rewritten query $\rightarrow$ Question + hypothetical passage	0.1040 [0.0740, 0.1340]	0.1057 [0.0791, 0.1364]	59 / 7	2.38×10^{-11}
Answer EM/F1	BM25 + Dense Hybrid $\rightarrow$ HyDE-style RAG	0.0440 [0.0160, 0.0740]	0.0529 [0.0264, 0.0803]	37 / 15	0.0032
Answer EM/F1	Rule-based Evidence-Expanded Iterative Retrieval $\rightarrow$ HyDE-style RAG	0.0480 [0.0220, 0.0740]	0.0540 [0.0296, 0.0788]	35 / 11	0.0005
Answer EM/F1	HyDE-style RAG $\rightarrow$ HyDE-style RAG + Conservative Verifier	0.0040 [0.0000, 0.0100]	0.0030 [0.0000, 0.0080]	2 / 0	0.5000
Retrieval All/Recall	Dense RAG $\rightarrow$ HyDE-style RAG	0.2380 [0.2000, 0.2780]	0.1200 [0.1010, 0.1410]	120 / 1	< 0.0001
Retrieval All/Recall	Single-query Reformulation RAG $\rightarrow$ HyDE-style RAG	0.2500 [0.2140, 0.2860]	0.1290 [0.1090, 0.1500]	125 / 0	< 0.0001
Retrieval All/Recall	Question + rewritten query $\rightarrow$ Question + hypothetical passage	0.2400 [0.2020, 0.2780]	0.1250 [0.1060, 0.1470]	123 / 3	< 0.0001
Retrieval All/Recall	Hybrid RAG $\rightarrow$ HyDE-style RAG	0.1200 [0.0840, 0.1560]	0.0590 [0.0410, 0.0780]	76 / 16	< 0.0001
Retrieval All/Recall	Rule-based Evidence-Expanded Iterative Retrieval $\rightarrow$ HyDE-style RAG	0.1140 [0.0780, 0.1480]	0.0580 [0.0390, 0.0780]	72 / 15	< 0.0001

All rows use the same 500 paired HotpotQA examples. For answer rows, metric 1 is Δ EM, metric 2 is Δ F1, $W \rightarrow C / C \rightarrow W$ count exact-match answer transitions, and p is the exact McNemar test over discordant EM outcomes. For retrieval rows, metric 1 is Δ all-support hit@10, metric 2 is Δ supporting-title recall@10, $W \rightarrow C / C \rightarrow W$ count target-only and baseline-only full-support retrieval, and p is the exact McNemar test over full-support retrieval transitions. Intervals are paired bootstrap 95% percentile intervals over per-example deltas.

S2 Full HotpotQA Paired Statistics

The HotpotQA paired statistics combine answer-side paired bootstrap/McNemar checks with retrieval-side paired checks over the same 500 examples. This table is the main supplemental lookup for the HotpotQA reliability claims in the manuscript.

Table S4: 2WikiMultihopQA paired statistics and secondary absolute results.

Panel A: absolute retrieval-reader results.

Method	Any hit@10	All hit@10	Recall@10	EM	F1
One-step Dense RAG	0.9840	0.3580	0.6695	0.5400	0.5767
BM25 RAG	0.9980	0.4320	0.7250	0.5840	0.6169
BM25 + Dense Hybrid	0.9980	0.4480	0.7330	0.5840	0.6220
HyDE-style RAG	0.9940	0.6680	0.8575	0.6620	0.7289

Panel B: paired HyDE-over-baseline diagnostics.

Baseline → target	Δ EM [95% CI]	Δ F1 [95% CI]	EM W→C / C→W	Δ All@10 [95% CI]	Δ Recall@10 [95% CI]	All@10 W→C / C→W
Dense RAG → HyDE-style RAG	0.1220 [0.0900, 0.1580]	0.1523 [0.1186, 0.1881]	72 / 11	0.3100 [0.2680, 0.3520]	0.1880 [0.1665, 0.2085]	157 / 2
BM25 RAG → HyDE-style RAG	0.0780 [0.0480, 0.1120]	0.1121 [0.0793, 0.1450]	54 / 15	0.2360 [0.1940, 0.2780]	0.1325 [0.1115, 0.1550]	133 / 15
BM25 + Dense Hybrid → HyDE-style RAG	0.0780 [0.0480, 0.1080]	0.1069 [0.0751, 0.1399]	53 / 14	0.2200 [0.1800, 0.2620]	0.1245 [0.1020, 0.1475]	125 / 15

The 2WikiMultihopQA split is used as a secondary generalization check rather than as the main benchmark. Panel B uses the same 500 paired examples and frozen reader/retrieval artifacts. Answer-side intervals are paired bootstrap intervals over EM and F1 deltas; retrieval-side intervals are paired bootstrap intervals over all-support hit@10 and supporting-title recall@10 deltas. W→C/C→W count exact-match answer transitions for answer rows and target-only/baseline-only full-support retrieval for retrieval rows.

S3 2Wiki Paired Statistics

The 2WikiMultihopQA split is reported as a secondary generalization check under the same processed local-corpus protocol, reader prompt, top- $k = 10$ retrieval budget, and local candidate-pool setting. The paired rows reuse frozen answer and retrieval artifacts and are interpreted as secondary diagnostics.

Table S5: Retrieval-side sensitivity to exact title–paragraph deduplication in the local joint index.

Dataset	Method	Retain all@10	Dedup all@10	Δ all	Retain recall	Dedup recall	Δ recall
HotpotQA	Dense	0.6300	0.6320	+0.0020	0.8080	0.8090	+0.0010
HotpotQA	Hybrid	0.7480	0.7520	+0.0040	0.8690	0.8710	+0.0020
HotpotQA	HyDE	0.8680	0.8680	+0.0000	0.9280	0.9280	+0.0000
2WikiMultihopQA	Dense	0.3580	0.3920	+0.0340	0.6695	0.6920	+0.0225
2WikiMultihopQA	Hybrid	0.4480	0.4600	+0.0120	0.7330	0.7415	+0.0085
2WikiMultihopQA	HyDE	0.6680	0.6760	+0.0080	0.8575	0.8625	+0.0050

Retain duplicates uses the canonical joint local index with repeated title–paragraph records preserved. Dedup removes exact normalized title–paragraph duplicates before retrieval, then rebuilds dense, BM25, hybrid, and HyDE retrieval artifacts with the same top- $k = 10$ setting. Metrics still collapse duplicate supporting titles during evaluation. Reader answers are not regenerated for the deduplicated indexes, so EM/F1 are not reported in this table. The CSV artifact additionally reports average duplicate title slots and exact duplicate slots in the retrieved top-10 lists.

S4 Deduplication Sensitivity

The local corpus for each dataset is the union of the released question-level candidate pools, indexed jointly within the dataset. Retrieval representations encode document titles and paragraph texts only. Question identifiers, supporting-evidence flags, gold answers, and evaluation metadata are retained for analysis but are not encoded by dense, BM25, or hybrid retrieval. Exact duplicate title–paragraph records are retained before indexing; if a paragraph appears in multiple question-level pools, each occurrence remains a separate document with a source-question-specific document identifier. Supporting-title metrics collapse duplicate retrieved titles before computing any-hit@10, all-hit@10, and supporting-title recall@10.

The sensitivity table shows that exact title–paragraph deduplication leaves the method ordering unchanged. HotpotQA HyDE metrics remain unchanged, while all three tested methods improve slightly on 2WikiMultihopQA where repeated local-pool records occupy more top-10 slots. Reader answers are not regenerated for the deduplicated indexes, so this analysis should be read as a retrieval-side robustness check rather than an EM/F1 robustness check. We therefore keep the retained-duplicate joint index as the canonical protocol because it mirrors the released local-pool artifacts, and treat deduplication as a robustness check rather than a protocol change.

Table S6: HyDE verifier guard ablation reconstructed from existing verifier outputs.

Variant	N	Verified	Changed	EM	F1	$\Delta F1$	$W \rightarrow C / C \rightarrow W$
HyDE reader only	500	0	0	0.6840	0.8105	0.0000	0 / 0
HyDE + verifier raw	500	120	5	0.6840	0.8119	0.0013	2 / 2
HyDE + no-abstention guard	500	120	5	0.6840	0.8119	0.0013	2 / 2
HyDE + numeric-unit guard	500	120	3	0.6880	0.8135	0.0030	2 / 0
HyDE + both guards	500	120	3	0.6880	0.8135	0.0030	2 / 0

All rows reuse the same HyDE reader answers and the same 120 risk-selected verifier outputs. The raw row applies verifier final answers directly. The final row applies both deterministic guards: block new abstentions and block unsupported numeric-unit additions. The ablation is intended to justify the verifier’s safety boundary, not to present verification as the main performance driver.

S5 Guarded Answer Canonicalization Decomposition

The verifier outputs are analyzed as a HyDE-native guarded answer-canonicalization layer. The decomposition below reuses the same HyDE reader answers and the same 120 risk-selected verifier outputs, then reconstructs raw merging and guarded merging policies without new model calls.

S5.1 Guarded Merge Rule

Algorithm S1: Guarded canonicalization inference rule.

1. The reader receives the original question q and retrieved corpus evidence E and returns an initial short answer a_0 .
2. The rule-based selector computes a risk score $s(q, a_0)$. If $s(q, a_0) = 0$, set the final answer $a_f = a_0$ and stop.
3. If $s(q, a_0) > 0$, query the conservative verifier with (q, E, a_0) and parse its JSON output for the proposed final answer $\hat{a}$. If JSON parsing fails or the required answer field is missing, set $a_f = a_0$ and stop.
4. Apply the abstention guard: if a_0 is not an abstention but $\hat{a}$ is an abstention, set $a_f = a_0$ and stop.
5. Apply the numeric-unit guard: if a_0 is a bare number and $\hat{a}$ only appends a high-risk unit suffix, set $a_f = a_0$ unless the multi-word unit is explicitly requested by the question.
6. Otherwise set $a_f = \hat{a}$.

The final answer is therefore either the original reader answer or a selected-and-guarded verifier output for risk cases. The guard-decomposition table above reports the effect of raw merging and each deterministic guard using the same frozen verifier outputs.

Table S7: Full structural call and retrieval-query accounting on the released IRCot HotpotQA test_subsampled split.

Method	Q-LLM/q	Ret.-q/q	Reader/q	Verifier/q	Total LLM/q	Prompt chars/q	Evidence chars/q
LLM-only / No Retrieval	0.00	0.00	1.00	0.00	1.00	368	0
One-step Dense RAG	0.00	1.00	1.00	0.00	1.00	6010	5149
Multi-query RAG	0.00	3.00	1.00	0.00	1.00	6010	5149
Single-query Reformulation RAG	1.00	1.00	1.00	0.00	2.00	6552	5169
BM25 RAG	0.00	1.00	1.00	0.00	1.00	6156	5295
BM25 + Dense Hybrid	0.00	2.00	1.00	0.00	1.00	5996	5135
Rule-based Evidence-Expanded Iterative Retrieval	0.00	4.00	1.00	0.00	1.00	6065	5204
HyDE-style RAG	1.00	1.00	1.00	0.00	2.00	6508	5121
HyDE-style RAG + Guarded Canonicalization	1.00	1.00	1.00	0.24	2.24	8214	5121

Q-LLM/q counts query-side generation calls for matched-call reformulation or HyDE-style hypothetical-passage generation. Ret.-q/q counts retrieval query executions per question: Hybrid executes one dense and one BM25 retrieval, Multi-query executes three dense query variants, and Rule-based Evidence-Expanded Iterative Retrieval executes one first-round dense query plus three evidence-expanded dense queries. Total LLM/q counts query-side generation, reader answering, and selective verifier-backed canonicalization calls; it does not include retrieval query executions. Prompt chars/q reports average prompt characters per question from actual JSONL prompt files. Evidence chars/q reports serialized reader evidence length. The current logs do not store provider-side token usage, wall-clock latency, or billing. No method uses task-specific fine-tuning or model training.

S6 Full Call and Prompt-Character Accounting

The main manuscript reports a compact LLM-call profile. The full accounting below separates query-side LLM calls, retrieval query executions, reader calls, verifier calls, total LLM calls, prompt-character proxies, evidence-character averages, and whether any method uses task-specific training. These are artifact-derived prompt-character and call-count statistics, not provider-side token, latency, or billing logs.

Table S8: Implemented baseline and control retrieval parameters used in the frozen HotpotQA artifacts.

Method	Query or document construction	Ranking and fusion rule	Frozen parameters and artifact notes
BM25 RAG	Query tokens are extracted from the original question by lowercasing and matching [a-z0-9]+. Each corpus record is represented as <code>title title text</code> , so the title receives a two-count lexical boost and the paragraph body appears once.	Standard BM25 lexical scoring with Robertson-style IDF, term-frequency saturation, and document-length normalization. No stopwords removal, stemming, phrase matching, synonym expansion, or learned reranking is applied.	$k_1 = 1.5$, $b = 0.75$, $\text{top-}k = 10$. The retained-duplicate local index is used for the canonical table rows; the deduplicated variant is reported only in Table S5.
BM25 + Dense Hybrid	The dense input list is the one-step all-MiniLM-L6-v2 top-10 list; the lexical input list is the BM25 top-10 list above. Both lists contain title and paragraph text only, with analysis metadata excluded from retrieval representations.	Reciprocal-rank fusion over the two ranked lists: $\text{score}(d) = I[d \in D]/(60 + r_D(d)) + I[d \in B]/(60 + r_B(d))$. Missing-list contributions are zero; fused documents are sorted by this score.	Dense weight = 1.0, BM25 weight = 1.0, RRF constant = 60, dense candidates = 10, BM25 candidates = 10, final top- $k = 10$. No dense/BM25 raw-score calibration is used.
Multi-query RAG	Three deterministic query variants are built from the question: the original question, a capitalized-or-numeric entity-term query, and a non-stopword keyword query. A punctuation-normalized fallback is used only if fewer than three distinct variants are available. No LLM is used.	Each query is encoded with the same dense encoder and retrieves 10 candidates. Candidate lists are fused by weighted reciprocal-rank fusion, with query-index weights 0.5 ^j in the frozen HotpotQA artifact and RRF constant 60; ties are broken by the best weighted dense similarity stored for the candidate.	Number of queries = 3, per-query candidates = 10, final top- $k = 10$, <code>query_decay</code> = 0.5. The frozen artifact has the same top-10 document set as one-step Dense RAG for all 500 HotpotQA examples, but only 39/500 examples have the same order; reader context order can therefore change without adding new documents.
Rule-based Evidence-Expanded Iterative Retrieval	The first stage is the one-step dense top-10 retrieval. The second stage constructs four dense queries: the original question plus one expanded query for each of the first three first-round documents, formatted as the question, the related evidence title, and up to 700 characters of related evidence text. No reader output and no LLM call are used to form these queries.	The four dense queries each retrieve 10 candidates. Candidate documents are merged by document ID; each candidate keeps its maximum adjusted dense similarity, with weight 1.0 for the original query and 0.97 for evidence-expanded queries. The final list is sorted by the retained adjusted score.	First-round expansion documents = 3, per-query candidates = 10, retrieval query executions per question = 4, final top- $k = 10$. This is a fixed heuristic iterative control rather than a reproduction of IRCoT. The final top-10 is a re-ranked union of the second-stage candidate set, not a simple append to the first-stage list.

These rows describe the implemented retrieval baselines and controls used for the reported HotpotQA comparisons. Dense retrieval, Multi-query retrieval, Rule-based Evidence-Expanded Iterative Retrieval, and HyDE-style retrieval all use the same `sentence-transformers/all-MiniLM-L6-v2` encoder and normalized FAISS inner-product search unless a row states otherwise. Multi-query and Rule-based Evidence-Expanded Iterative Retrieval are fixed heuristic controls, and Single-query Reformulation is a matched-call mechanism control; they are not claimed as faithful public-method reproductions. The matched-call Single-query Reformulation and HyDE-style query prompts are specified in Section S7.

S7 Prompt and API Configuration

This section gives the complete prompt templates used by the canonical query-side generation, matched-call query reformulation, equal-budget query-form diagnostic, evidence-grounded short-answer reader, and conservative verifier. The reader prompt uses generic short-answer formatting instructions for common multi-hop QA answer types and is applied unchanged to HotpotQA and 2WikiMultihopQA. The reader and verifier evidence blocks serialize the top-10 retrieved passages with title and paragraph text, truncated to at most 900 characters per document.

S7.1 HyDE Hypothetical-Passage Generation Prompt

You are generating a hypothetical supporting passage for retrieval.
Given the question below, write a concise factual-sounding supporting passage that would likely contain the answer.
The passage may be imperfect, but it should include key entities, relations, and context useful for retrieving evidence.
Do not explain the task. Do not include bullet points. Write one short paragraph.

Question:
{question}

Hypothetical supporting passage:

S7.2 Matched-Call Single-Query Reformulation Prompt

You are rewriting a multi-hop question into a single retrieval query.
Given the question below, produce one concise search query for dense retrieval.
Use only entities, relations, and constraints already present in the question.
Do not answer the question. Do not add facts that are not stated in the question.
Do not write a paragraph or explanation. Output only the single retrieval query.

Question:
{question}

Single retrieval query:

S7.3 Equal-Budget Query-Form Prompts and Serialization

The equal-budget diagnostic reported in main Tables 5 and 6 uses four separate query-form prompts with a shared wrapper. Each prompt asks for retrieval text, not an answer, uses one query-side generation call, targets 35–45 English words around a 40-word target, and sets the generation budget to `max_tokens=64` with temperature 0. The prompt rows store gold answers and supporting-fact metadata only for later scoring; these fields are not inserted into the user prompt. We did not discard, truncate, or regenerate outputs solely because they fell outside the requested word window; the realized word counts are reported with the retrieval summary. Failed API requests are retried by the generation wrapper and, under `--resume`, rerun only as missing rows rather than replaced by hand-written text.

The generated text is used for retrieval only. For all four equal-budget rows, the BGE-base encoder input is the original question plus the generated retrieval text:

{question}

Generated retrieval text:
{generated_text}

Thus, the document-like condition is encoded as $[q; G_{\text{doc}}(q)]$, not as $G_{\text{doc}}(q)$ alone. The exact separator is two newline characters, the label “Generated retrieval text:”, and one newline before the model output. The BGE-base tokenizer then encodes this single serialized string with a 512-token query limit and right truncation. Lists, semicolons, bullets, and sentences are controlled only by the prompt text below: the keyword mode requests comma-separated phrases rather than sentences, the direct-rewrite mode requests one sentence, the decomposition mode requests numbered subquestions, and the document-like mode requests prose rather than bullets. The prompts do not include retrieved evidence or corpus text. They also do not impose a global ban on adding likely bridge concepts: the keyword/entity and document-like variants explicitly invite likely bridge concepts or plausible supporting context, while the direct-rewrite variant asks only to preserve the original intent and make implicit relations explicit.

Keyword/Entity Expansion

You are generating retrieval text for a controlled multi-hop QA query-composition diagnostic. Write a compact keyword/entity list for dense retrieval. Include entities, relations, constraints, and likely bridge concepts. Use comma-separated phrases, not sentences.

Use only one generation call. Do not answer the question. Do not explain the task.
Aim for 35-45 English words. Prefer concrete entities and relations over filler.
Output only the retrieval text.

Question:
{question}

Retrieval text:

Direct Rewrite

You are generating retrieval text for a controlled multi-hop QA query-composition diagnostic. Write one single dense-retrieval rewrite of the question. Preserve the original intent, make implicit relations explicit, and keep it as one sentence.

Use only one generation call. Do not answer the question. Do not explain the task.
Aim for 35-45 English words. Prefer concrete entities and relations over filler.
Output only the retrieval text.

Question:
{question}

Retrieval text:

Question Decomposition

You are generating retrieval text for a controlled multi-hop QA query-composition diagnostic. Write numbered subquestions that decompose the multi-hop question into retrieval steps. Keep the total output within the same length budget.

Use only one generation call. Do not answer the question. Do not explain the task.
Aim for 35-45 English words. Prefer concrete entities and relations over filler.
Output only the retrieval text.

Question:
{question}

Retrieval text:

Document-Like Passage

You are generating retrieval text for a controlled multi-hop QA query-composition diagnostic. Write one short encyclopedia-style passage that could plausibly support answering the question. Use prose rather than bullets.

Use only one generation call. Do not answer the question. Do not explain the task.
Aim for 35-45 English words. Prefer concrete entities and relations over filler.
Output only the retrieval text.

Question:
{question}

Retrieval text:

S7.4 Evidence-Grounded Short-Answer Reader Prompt

The frozen prompt text below uses the historical word “extractive” in its system role. In this manuscript, this denotes evidence-constrained short-answer reading rather than standard span-extractive QA: the answer is not required to be a contiguous span copied from one evidence passage.

You are an extractive multi-hop question answering system.
Use only the evidence below. Combine facts across evidence items when needed.
Return the exact short answer phrase requested by the question.

Important rules:

- Return only the answer, with no explanation.
- Prefer the most specific answer phrase in the evidence.
- For yes/no questions, answer only "yes" or "no".
- For location questions, include the full location if the evidence gives it.
- For date/age comparison questions, the earlier birth date means the person is older.
- For role/position questions, return the exact position title, not a broader category.
- Do not answer "I don't know" if the evidence contains enough information to infer the answer.

Question:

{question}

Evidence:

{evidence}

Exact short answer:

S7.5 Conservative Verification Prompt

You are a conservative evidence-grounded answer verification agent.

Your default behavior is to KEEP the initial answer. Rewrite it only when there is a clear benchmark-style formatting issue and the evidence directly supports the corrected phrase.

Allowed corrections:

1. Nationality questions: prefer adjectival nationality when the evidence supports it.
Example: "American" instead of "United States citizen".
2. Count questions: if the initial answer is only a number but the question asks for a counted object, include the unit from the evidence. Example:
"12 member universities" instead of "12".
3. Date questions: if the question asks for an event time and the evidence gives a more complete month/year phrase, include the complete phrase.
4. Yes/no questions: if the question asks a yes/no comparison and the evidence directly supports it, normalize the answer to "Yes" or "No".
5. Abstained answers: if the initial answer is "I don't know" but the evidence directly contains the exact requested short answer, provide that answer.
Otherwise keep "I don't know".
6. Location, role, title, or type questions: only add a missing qualifier when the question asks for that qualifier and the exact phrase appears in evidence.

Do not make these changes:

1. Do not abstain if the initial answer is directly supported by any evidence.
2. Do not expand a familiar short person name to a full legal name unless the question explicitly asks for the full name.
3. Do not remove location qualifiers such as state/country names.
4. Do not remove parenthetical title explanations.
5. Do not rewrite a correct answer merely because another phrase is also present.
6. Do not add units to a numeric answer when the question only asks for a number.
7. Do not replace a precise answer with a longer phrase unless the longer phrase is explicitly requested by the question.

Return only valid JSON with this exact schema:

```
{"verdict": "keep|correct|abstain", "final_answer": "...", "reason": "short reason"}
```

Question:

{question}

Initial answer:

{prediction}

Evidence:

{evidence}

JSON:

Table S9: Model invocation ledger for frozen hosted-Qwen artifacts.

Artifact group	Local freeze window	Stored model field(s)	Calls / rows	Frozen artifact source
HotpotQA retrieval baselines	2026-07-02 to 2026-07-04	qwen3.7-max	5 × 500	LLM-only, Dense, BM25, Hybrid, and Multi-query answer JSONL files.
HotpotQA matched-call query reformulation	2026-07-08 to 2026-07-09	qwen3.7-max	3 × 500	Single-query reformulation generation, rewritten-query reader, and question + rewritten-query reader JSONL files.
HotpotQA HyDE and iterative runs	2026-07-04 to 2026-07-06	qwen3.7-max	4 × 500	HyDE generation, HyDE reader, Rule-based Evidence-Expanded Iterative Retrieval reader, and HyDE hypothetical-only mechanism answer files.
HotpotQA conservative verifier	2026-07-04	qwen3.7-max	120 verifier calls; 500 merged rows	Risk-selected verifier answer JSONL plus numeric-guarded merged evaluation file.
2WikiMultihopQA secondary runs	2026-07-05 to 2026-07-06	qwen3.7-max	5 × 500	2Wiki Dense, BM25, Hybrid, HyDE generation, and HyDE reader answer files.
Equal-budget query-form generation	2026-07-21 to 2026-07-23	qwen3.7-max	8 × 500	Post-hoc HotpotQA and 2Wiki keyword/entity, direct-rewrite, question-decomposition, and document-like generation files.
Equal-budget Qwen-Turbo reader check	2026-07-23 to 2026-07-25	qwen-turbo	6 × 500	Direct-rewrite, keyword/entity-expansion, and document-like top-10 reader answer files for HotpotQA and 2Wiki.

All rows use the Alibaba Cloud Bailian/DashScope OpenAI-compatible chat-completions API wrapper with temperature=0 decoding (temperature=0.0). The primary pipeline uses the stored API model string `qwen3.7-max`; the equal-budget query-form generation reuses `qwen3.7-max` as a post-hoc diagnostic; the targeted cross-reader check intentionally uses `qwen-turbo`. The configured `max_tokens` budgets are reported separately as 64 for HyDE passage generation, equal-budget query-form generation, rewritten-query generation, reader answers, and verifier JSON output. Temperature 0 does not guarantee byte-identical replay across requests or hosted-model alias updates. Freeze windows are local artifact timestamps, not provider request logs. Table S10 records the API-level reproducibility boundary, and the complete artifact-level ledger is recorded in `docs/model_invocation_ledger.md`.

S7.6 API, Model Invocation, and Result Ledger

The hosted model is recorded at the artifact level because the provider exposes a hosted alias rather than a public immutable checkpoint. Reported reader, query-generation, and verifier artifacts use the stored API model string `qwen3.7-max`. The following tables freeze the local invocation windows, stored model fields, prompt hashes, retry policy, parsing policy, and raw-output boundaries.

For the structured verifier output, we audited the canonical HotpotQA HyDE verifier artifact listed in Table S9. All 120 verifier outputs were successfully parsed as direct JSON under the reported 64-token budget. No substring JSON-recovery fallback was needed, no empty verifier output was observed, and no final JSON parse failure remained in the frozen verifier artifact. The generation wrapper does not store provider-side `finish_reason` or per-request retry counters, so truncation is bounded here by artifact-level parse success rather than by provider response metadata.

S7.7 Dense Encoder and Truncation Configuration

All canonical MiniLM-based dense retrieval variants use `sentence-transformers/all-MiniLM-L6-v2` through the `SentenceTransformer.encode` interface with normalized embeddings and FAISS inner-product search. The local audit environment reports `sentence-transformers==5.6.0`; the local MiniLM model cache records a `sentence_bert_config.json` value of `max_seq_length=256`. The underlying uncased BERT tokenizer uses WordPiece tokenization and the default Transformers truncation side is right truncation. Thus, MiniLM dense inputs are encoded as a single sequence with a shared 256-WordPiece-token budget, including the tokenizer’s special tokens, and excess content is dropped from the right after serialization. The retained HotpotQA and 2Wiki MiniLM indexes are FAISS `IndexFlatIP` indexes over normalized 384-dimensional embeddings, so inner-product search is cosine-style nearest-neighbour retrieval.

The same tokenizer, 256-token budget, and right-truncation rule are applied to all dense query-composition diagnostic

Table S10: Hosted Qwen API reproducibility record for frozen generation artifacts.

Field	Frozen record or limitation
Provider and interface	Alibaba Cloud Bailian/DashScope OpenAI-compatible chat-completions interface, accessed through the local <code>LLM_BASE_URL</code> environment variable. API credentials and endpoint URL are not stored in the artifact release.
API model string	Reported primary reader, query-generation, and verifier artifacts use the hosted API model string <code>qwen3.7-max</code> . The post-hoc equal-budget query-form generation also uses <code>qwen3.7-max</code> , while the targeted equal-budget reader robustness check uses <code>qwen-turbo</code> as a second hosted reader and is reported separately. The model string is stored in the local JSONL artifacts when the generation wrapper writes model metadata.
Snapshot/version	No immutable provider-side snapshot identifier was available for the hosted <code>qwen3.7-max</code> alias. Canonical runs are therefore frozen by local prompt and output artifacts rather than by a public model checkpoint.
Module model-string consistency	Primary HyDE generation, matched-call query reformulation, reader, 2Wiki, and verifier runs use the same API model string, <code>qwen3.7-max</code> . The post-hoc equal-budget query-form generation also uses <code>qwen3.7-max</code> but is not part of the primary frozen scoring window. The Qwen-Turbo check intentionally changes only the reader over fixed equal-budget retrieval evidence. Because these are hosted aliases rather than immutable snapshots, the paper does not claim replay against the exact same provider-internal model state after an alias evolves.
Experiment window	Primary Qwen3.7-Max artifacts were generated or frozen between 2026-07-02 and 2026-07-09 local time. Post-hoc equal-budget Qwen3.7-Max query-form generations were frozen between 2026-07-21 and 2026-07-23 local time. The targeted Qwen-Turbo fixed-evidence reader artifacts were frozen between 2026-07-23 and 2026-07-25 local time. Artifact-level windows and paths are listed in Table S9.
Endpoint region/type	Endpoint type is OpenAI-compatible /v1 chat completions. Provider region was not exported into the local JSONL artifacts and is therefore not claimed as a reproducible field.
Temperature and output budgets	<code>temperature=0.0</code> . Configured <code>max_tokens</code> : HyDE passage generation = 64; rewritten-query generation = 64; reader answer generation = 64; verifier JSON generation = 64. Temperature 0 is used to reduce sampling variation, but byte-identical replay is not claimed for a hosted alias.
<code>top_p</code> , seed, and thinking mode	No <code>top_p</code> , seed, logprob, response-format, or reasoning/thinking-mode parameter is explicitly sent by the wrapper, so provider defaults apply where relevant. Seed-controlled replay and thinking-mode equivalence are therefore not claimed.
Retry and timeout policy	Timeout is 60 seconds by default. Each request is attempted up to 3 times with 5 seconds between retries. If all retries fail, the script raises the final exception rather than writing a synthetic answer row. The <code>--resume</code> option skips completed IDs already present in the output JSONL and appends only missing rows.
API failure replacement policy	Reported <code>qwen3.7-max</code> HyDE, matched-call, 2Wiki, reader, and verifier runs, as well as the <code>qwen-turbo</code> fixed-evidence reader check, use completed rows from the frozen output files. Failed requests are rerun only as missing rows under <code>--resume</code> ; they are not silently replaced by hand-written answers.
Prompt version/hash	Prompt-builder script SHA-256 prefixes: HyDE 89ECB4FB, reader 98904C37, query reformulation FF9EA7EF, equal-budget query-form A234A00B, verifier BB2495BC. Canonical HotpotQA prompt-file SHA-256 prefixes: HyDE generation 4D43047C, HyDE reader DCBC0B3E, verifier risk-120 763D188C. Equal-budget prompt-file SHA-256 prefixes: HotpotQA keyword 048B13BE, direct 2777A456, decomposition 4244BD3E, document-like FF4A4ECF; 2Wiki keyword EC679B8B, direct E7464644, decomposition BF51AE1A, document-like CF15FFFF.
Malformed JSON handling	Reader, HyDE, and query-reformulation generations are stored as raw text predictions and are not JSON-parsed. Verifier outputs are parsed as JSON when possible, with substring JSON extraction for wrapped responses. If verifier parsing fails, or if the parsed object lacks a nonempty <code>final_answer</code> field, the fail-safe policy retains the initial reader answer <code>a0</code> ; malformed verifier text is never used as a candidate final answer. No automatic re-request is triggered solely by malformed JSON or JSON parse failure.
Verifier JSON parse audit	The canonical HotpotQA HyDE verifier artifact contains 120 verifier rows generated with <code>max_tokens=64</code> . All 120 <code>prediction</code> fields were successfully parsed as direct JSON objects; substring fallback was used for 0 rows; final JSON parse failures were 0; empty verifier outputs were 0. All 120 verifier rows store the model field <code>qwen3.7-max</code> . The wrapper does not retain provider <code>finish_reason</code> fields or per-request retry counters, and no automatic re-request rule is triggered solely by suspected output truncation.
Raw request/response retention	Local artifacts retain the full user prompt text, retrieved evidence, prediction text, model field, gold-answer metadata for scoring, and supporting-fact metadata. They do not retain the provider-side raw request envelope, raw response envelope, request IDs, token accounting, latency, billing, or per-request provider timestamps. Local artifact modification times provide freeze timestamps in the ledger.

Because the hosted model alias may evolve, all reported generations are accompanied by frozen raw prompts, local raw text outputs, local artifact freeze timestamps, and stored model strings. This table supports artifact-level reproducibility and rescoring, not exact replay of a provider-internal model snapshot or provider-side request log reconstruction.

rows on HotpotQA: question only (q), single rewritten query (r), question plus rewritten query ($q + r$), hypothetical passage only (h), and question plus hypothetical passage ($q + h$). The generated text is character-capped before this tokenizer step: reformulated queries use at most 300 characters, and HyDE hypothetical passages use at most 900 characters with an ellipsis appended if capped. The combined rows share one encoder input sequence rather than separate per-field budgets for the question and generated text.

Dense query serialization is fixed by the retrieval scripts. The $q + h$ row serializes the original question, two newline characters, the label “Hypothetical supporting passage:”, one newline, and the generated passage. The $q + r$ row uses the same layout with the label “Rewritten retrieval query:”. The q , r , and h rows encode only their corresponding serialized text. Corpus documents are serialized for dense indexing as `{title}. {paragraph}` and use the same 256-token right-truncation rule; there is no separate title or paragraph token budget. Question identifiers, support flags, answers, and evaluation metadata are not encoded.

All post-hoc BGE-base diagnostics use one frozen encoder configuration across the rows being compared:

- model identifier: `BAAI/bge-base-en-v1.5`;
- local Hugging Face main revision: `a5beb1e3e68b9ab74eb54cfd186867f64f240e1a`;
- local `model.safetensors` SHA-256 prefix: `C7C1988AAE20`;

- local config.json SHA-256 prefix: BC00AF31A4A3;
- local tokenizer_config.json SHA-256 prefix: 9261E7D79B44;
- implementation: transformers.AutoTokenizer and AutoModel, not SentenceTransformer.encode; the local audit environment reports transformers==4.57.6 and torch==2.8.0.

The BGE diagnostics use CLS pooling, taking `last_hidden_state[:, 0]` as the sequence embedding. Corpus-scale scripts expose a mean-pooling option for sensitivity work, but the reported BGE rows use the default `pooling=cls`. Both document and query inputs use `max_length=512`, tokenizer-side right truncation, and identical L2 normalization with `F.normalize(..., p=2, dim=1)`; this is equivalent to using normalized embeddings. No query instruction prefix was added. Queries and documents are encoded as plain serialized strings, without BGE’s recommended instruction-style query prefix.

BGE document serialization is `{title}. {paragraph_text}` with no separate title budget. The stronger-encoder query-composition diagnostic serializes question-only as *q*, single rewrite as *r*, hypothetical-only as *h*, and uses the following combined-query strings:

```
{question}\n\nRewritten retrieval query:\n{rewrite}
```

```
{question}\n\nHypothetical supporting passage:\n{hypothetical passage}
```

The equal-budget diagnostic serializes each generated form identically as:

```
{question}\n\nGenerated retrieval text:\n{generated_text}
```

Thus the document-like condition encodes $[q; G_{\text{doc}}(q)]$ rather than $G_{\text{doc}}(q)$ alone. BGE corpus-scale and hard-negative diagnostics use the same question-only, single-rewrite, and question-plus-hypothetical serializations as the stronger-encoder diagnostic. BGE diagnostics do not build a FAISS index: they compute dense score matrices as the dot product between L2-normalized query and document embeddings and select top-*k* with `torch.topk`, making the score equivalent to cosine similarity.

S8 Corpus-Scale Retrieval Stress Test

The main experiments use processed local candidate pools. To test sensitivity to a substantially noisier index without claiming full-wiki retrieval, we compare the original retained-duplicate local index with expanded 50k and 100k stress-test indexes. The expanded corpora preserve the released evaluation corpus and then add exact-title-paragraph-deduplicated distractor passages sampled from the corresponding IRCot processed training split using seed 20260721. The expansion is gold-preserving by construction because the original evaluation support passages remain in the index; it is therefore a diagnostic retrieval stress test rather than a deployable retrieval protocol.

All rows use BAAI/bge-base-en-v1.5, CLS pooling, 512-token dense inputs, top-*k* = 10, and the frozen question-only, single-query reformulation, and HyDE hypothetical-passage artifacts. No reader calls are rerun for this stress test. The script is `src/evaluation/analyze_corpus_scale_retrieval_stress.py`; the summary CSV, expanded corpora, retrieval JSONL files, and embedding caches are stored under `local-artifacts/corpus_scale_stress`. Paired bootstrap intervals are computed over per-example retrieval deltas.

Table S11: Full corpus-scale retrieval stress-test results. Local rows reuse the retained-duplicate BGE local-index artifacts from the main stronger-encoder reproduction table; only the 50k and 100k rows use newly expanded stress-test corpora.

Dataset	Docs	Query input	Any	All	Recall	HyDE delta vs. question	HyDE delta vs. rewrite
HotpotQA	Local	Question only	0.9940	0.8820	0.9380	–	–
HotpotQA	Local	Single rewritten query	0.9920	0.8340	0.9130	–	–
HotpotQA	Local	Question + hypothetical passage	0.9940	0.9200	0.9570	All +0.0380 [0.0100, 0.0660]; recall +0.0190 [0.0050, 0.0340]	All +0.0860 [0.0560, 0.1160]; recall +0.0440 [0.0280, 0.0590]
HotpotQA	50k	Question only	0.9820	0.7480	0.8650	–	–
HotpotQA	50k	Single rewritten query	0.9720	0.6520	0.8120	–	–
HotpotQA	50k	Question + hypothetical passage	0.9860	0.8560	0.9210	All +0.1080 [0.0680, 0.1460]; recall +0.0560 [0.0350, 0.0770]	All +0.2040 [0.1620, 0.2440]; recall +0.1090 [0.0860, 0.1320]
HotpotQA	100k	Question only	0.9800	0.6960	0.8380	–	–
HotpotQA	100k	Single rewritten query	0.9680	0.5960	0.7820	–	–
HotpotQA	100k	Question + hypothetical passage	0.9780	0.8340	0.9060	All +0.1380 [0.1000, 0.1780]; recall +0.0680 [0.0470, 0.0890]	All +0.2380 [0.1940, 0.2800]; recall +0.1240 [0.0990, 0.1480]
2Wiki	Local	Question only	0.9980	0.5100	0.7655	–	–
2Wiki	Local	Single rewritten query	1.0000	0.4720	0.7380	–	–
2Wiki	Local	Question + hypothetical passage	1.0000	0.7800	0.9130	All +0.2700 [0.2300, 0.3100]; recall +0.1475 [0.1290, 0.1675]	All +0.3080 [0.2680, 0.3520]; recall +0.1750 [0.1545, 0.1960]
2Wiki	50k	Question only	1.0000	0.4460	0.7270	–	–
2Wiki	50k	Single rewritten query	0.9960	0.4040	0.6900	–	–
2Wiki	50k	Question + hypothetical passage	0.9980	0.6900	0.8655	All +0.2440 [0.2040, 0.2840]; recall +0.1385 [0.1175, 0.1595]	All +0.2860 [0.2440, 0.3280]; recall +0.1755 [0.1545, 0.1980]
2Wiki	100k	Question only	1.0000	0.4180	0.7105	–	–
2Wiki	100k	Single rewritten query	0.9940	0.3600	0.6620	–	–
2Wiki	100k	Question + hypothetical passage	0.9980	0.6500	0.8440	All +0.2320 [0.1900, 0.2760]; recall +0.1335 [0.1110, 0.1560]	All +0.2900 [0.2460, 0.3340]; recall +0.1820 [0.1585, 0.2055]

Local denotes the original retained-duplicate local index used in the main stronger-encoder reproduction table (4,977 HotpotQA documents and 5,000 2Wiki documents). The rounded 5k stress-test cache is not used as a local-index replicate because the stress-construction script exact-deduplicates title–paragraph records before expansion and, for HotpotQA, pads the 4,977-document local corpus to 5,000 with train distractors. The 50k and 100k corpora preserve all released test-corpus records and add exact-title–paragraph-deduplicated contexts sampled from the corresponding IRCOT processed training split. All rows use BAAI/bge-base-en-v1.5, CLS pooling, 512-token dense inputs, top- $k = 10$, and the frozen generated query artifacts used in the stronger-encoder reproduction. Deltas are computed only for the HyDE query row and use paired bootstrap 95% confidence intervals over 500 per-example retrieval scores. The summary CSV is `local-artifacts/corpus_scale_stress/corpus_scale_retrieval_summary.csv`; Local retrieval JSONL files are the stronger-encoder reproduction artifacts in `local-artifacts/results`, while expanded-corpus retrieval JSONL files and embedding caches are stored under `local-artifacts/corpus_scale_stress`. This gold-preserving distractor expansion is a diagnostic stress test and should not be interpreted as full-wiki retrieval.

Table S12: Answer EM/F1 normalization and scoring rules.

Item	Rule used for reported answer metrics
Evaluator	Local HotpotQA-style reimplementation in <code>src/evaluation/evaluate_answers.py</code> ; SHA-256 prefix 55A65166A2FB. The official HotpotQA script is not imported at runtime, but the normalization steps match the standard HotpotQA answer-normalization recipe.
Shared across datasets	The same evaluator functions, <code>exact_match</code> and <code>f1_score</code> , are used for HotpotQA and 2WikiMultihopQA rows. No dataset-specific answer normalizer is used.
Lowercasing	Convert prediction and gold answer to Python strings and lowercase them before comparison.
Article removal	Remove standalone English articles <code>a</code> , <code>an</code> , and <code>the</code> using the regular expression <code>\b(a an the)\b</code> .
Punctuation	Delete ASCII punctuation characters from Python <code>string.punctuation</code> . Unicode punctuation is not separately canonicalized.
Whitespace	Split on whitespace and rejoin with a single ASCII space; leading, trailing, and repeated whitespace therefore collapse after lowercasing, article removal, and punctuation deletion.
Exact match	EM is 1 if the normalized prediction string is exactly equal to the normalized gold string, and 0 otherwise.
Token F1	Tokenize the normalized strings by whitespace. Let c be the multiset intersection count between predicted and gold tokens. If either side is empty and the other is nonempty, F1 is 0; if both are empty, F1 is 1. Otherwise precision is $c/ \text{pred} $, recall is $c/ \text{gold} $, and F1 is $2PR/(P + R)$.
Multiple gold answers	The released answer rows store one <code>gold_answer</code> string per example. The evaluator does not take a maximum over multiple aliases or alternative gold answers.
Empty predictions	Empty predictions are normalized to the empty string. They receive EM 1 and F1 1 only against an empty normalized gold answer; otherwise EM and F1 are 0.
Yes/no, numbers, dates	No special handling is applied beyond the same lowercasing, article removal, ASCII punctuation deletion, and whitespace collapse. Number words are not converted to digits, dates are not normalized, and yes/no answers are ordinary strings.

The same functions are imported by the answer-summary and paired-comparison scripts, including the BGE query-composition and equal-budget reader diagnostics.

S9 Answer and Evidence Metric Normalization

The answer scores reported as EM and token-level F1 use the same local evaluator for HotpotQA and 2WikiMultihopQA. This table defines answer scoring; it is separate from the answer-overlap diagnostics in Section S10, which compare generated query text against gold answers rather than scoring reader predictions.

Table S13: Supporting-title and supporting-paragraph metric normalization rules.

Item	Rule used for reported evidence metrics
Main supporting-title scoring	The reported any-hit@10, all-support hit@10, and supporting-title recall@10 use exact title strings from the retrieval artifact: <code>set(row["supporting_facts"]["title"])</code> for gold titles and <code>{doc["title"] for doc in row["retrieved"]}</code> for retrieved titles.
Title lowercasing and trimming	Main title metrics do not lowercase titles, strip titles, or delete ASCII/Unicode punctuation at scoring time. They rely on the released processed-data title strings and retrieved document title strings matching exactly.
Duplicate titles	Gold and retrieved titles are converted to Python sets before scoring, so repeated retrieved records with the same title count once. A row is all-support hit@10 when the nonempty gold-title set is a subset of the retrieved-title set; recall is the intersection size divided by the gold-title set size.
Evaluator paths and hashes	The canonical title evaluator is <code>src/evaluation/evaluate_retrieval.py</code> , SHA-256 prefix CDBB6B75E260. Equal-budget title contrasts use the same set rule in <code>src/evaluation/analyze_equal_budget_pairwise.py</code> , SHA-256 prefix 46AD7C79BC18; BGE retrieval-generation scripts store the same metrics during retrieval.
Paragraph-audit normalization	The supporting-paragraph audit uses <code>normalize_text(value) = " ".join(str(value or "").split()).casefold()</code> for both titles and paragraph texts, then forms exact keys as <code>(normalized_title, normalized_paragraph_text)</code> . It collapses whitespace and casefolds Unicode case, but it does not remove ASCII or Unicode punctuation.
Paragraph-audit gold keys	Gold paragraph keys are derived from corpus records with <code>is_supporting=true</code> and the matching <code>source_question_id</code> . Retrieved paragraph keys are built from the top-10 retrieved documents using the same normalized title-paragraph-text function.
Paragraph-audit duplicates	Paragraph keys, record ids, and titles are stored as sets. Duplicate title-paragraph records therefore collapse for paragraph-key scoring, while the optional record-id metric remains stricter because it requires the source-question-specific <code>doc_id</code> .
Paragraph-audit script	The audit script is <code>src/evaluation/analyze_supporting_paragraph_metrics.py</code> , SHA-256 prefix F8200E6D0911. Its generated table is Table S28.

Title-level metrics are the primary retrieval metrics in the main manuscript because the benchmark annotations are supporting-title based in the released processed rows. Paragraph-level metrics are a stricter artifact audit over the local corpus records, not a new sentence-level supporting-fact metric.

Table S14: Reproducibility rules for answer-string-overlap diagnostics.

Item	Rule used in the released audit scripts
Normalization	Convert to lowercase; remove standalone English articles <code>a</code> , <code>an</code> , and <code>the</code> ; remove ASCII punctuation using Python <code>string.punctuation</code> ; collapse consecutive whitespace. Punctuation is deleted rather than replaced, so hyphens, slashes, apostrophes, periods, and commas are removed under the same rule.
Overlap test	Let g be the normalized gold answer and z be the normalized generated query text: the HyDE hypothetical passage for HyDE rows or the rewritten query for the matched-call rewrite row. An overlap is counted when g is nonempty, z is nonempty, and g occurs as an exact substring of z .
Alias handling	No alias dictionary, entity linking, date parser, stemming, or paraphrase matching is used. Aliases count only if the normalized benchmark gold string itself appears in the generated query text.
Numbers and dates	No special numeric or date canonicalization is applied beyond punctuation removal and whitespace normalization. Number words are not mapped to digits, dates are not reordered, hyphenated dates lose their hyphens, and partial date aliases are not counted unless they satisfy the exact normalized substring rule.
Stop-answer or short-answer rule	A gold answer is flagged as short/ambiguous if its normalized form is <code>yes</code> , <code>no</code> , a pure numeric string, or a single token with at most three characters. This is the only stop-answer/minimum-length filter used for the non-trivial overlap count.
Non-trivial answer overlap	A non-trivial overlap is an overlap where the gold answer is not flagged as short/ambiguous. Thus the HotpotQA count of 252 is the subset of the 292 overlap cases after removing short/ambiguous answers.
Answer-absent subset	The answer-absent subset contains examples with no exact normalized overlap between the gold answer and the generated HyDE hypothetical passage. For HotpotQA, this gives $500 - 292 = 208$ examples; for 2WikiMultihopQA, this gives $500 - 347 = 153$ examples.

The scripts implementing these rules are `analyze_hyde_leakage.py`, `analyze_query_answer_overlap.py`, and `analyze_answer_absent_subset.py`. The rule is intentionally lexical: it is reproducible, but it does not rule out paraphrased answer content, aliases, model memory, or benchmark contamination.

Table S15: HyDE hypothetical-passage answer-overlap diagnostic on HotpotQA.

Group	N	All hit@10	Recall@10
All examples	500	0.8680	0.9280
Gold answer string appears	292	0.9452	0.9726
Gold answer string absent	208	0.7596	0.8654
Non-trivial answer string appears	252	0.9365	0.9683

A match is a normalized string diagnostic over generated hypothetical passages using the rules in Table S14. The hypothetical passage is used only as a retrieval query and is not included in the reader evidence prompt.

S10 Answer-Overlap Normalization

The following tables support the answer-string-overlap boundary and the secondary 2Wiki generalization analysis. The overlap diagnostics are query-side diagnostics: the hypothetical passage is used for retrieval, while the reader receives only the original question and retrieved corpus evidence.

The exact lexical implementation used by the released audit scripts is:

```
def norm(s):
    s = lower(str(s))
    s = remove_regex_word_articles(s, articles={"a", "an", "the"})
    s = delete_ascii_punctuation(s, chars=string.punctuation)
    return collapse_whitespace(s)

def is_short_or_ambiguous(gold):
    g = norm(gold)
    return (
        g in {"yes", "no"}
        or regex_fullmatch(r"\d+(?:\.\d+)?", g)
        or (len(split(g)) == 1 and len(g) <= 3)
    )

def has_answer_string_overlap(gold, generated_query):
    g = norm(gold)
    z = norm(generated_query)
    return bool(g) and bool(z) and g in z

def is_non_trivial_overlap(gold, generated_query):
    return has_answer_string_overlap(gold, generated_query) \
        and not is_short_or_ambiguous(gold)
```

No alias table, entity linker, date parser, stemming, paraphrase matcher, or numeric word-to-digit converter is applied before these functions.

The following gold-content diagnostics refine the exact-answer overlap analysis. They are not used for model selection or inference. The first table assigns each hypothetical passage to a mutually exclusive automatic stratum: exact answer string,

Table S16: Answer-overlap robustness analysis on HotpotQA.

Group	N	Dense all-hit	HyDE all-hit	Δ all-hit	Dense F1	HyDE F1	$W \rightarrow C / C \rightarrow W$
all examples	500	0.6300	0.8680	0.2380	0.7199	0.8105	53 / 8
answer in hyde	292	0.6644	0.9452	0.2808	0.7976	0.9085	33 / 0
answer not in hyde	208	0.5817	0.7596	0.1779	0.6110	0.6730	20 / 8
nontrivial answer in hyde	252	0.6825	0.9365	0.2540	0.8130	0.8953	22 / 0

Groups are defined by normalized string overlap between the gold answer and the generated hypothetical passage using the rules in Table S14. The answer-not-in-HyDE row is the robustness slice most relevant to answer-memory concerns; it uses existing answer files and does not require new LLM calls.

Table S17: Answer-absent subset paired analysis.

Dataset	N	Dense all-hit	HyDE all-hit	Dense recall	HyDE recall	Dense F1	HyDE F1	Δ F1 [95% CI]	$W \rightarrow C / C \rightarrow W$	McNemar p
HotpotQA	208	0.5817	0.7596	0.7812	0.8654	0.6110	0.6730	0.0620 [0.0170, 0.1080]	20 / 8	0.0357
2WikiMultihopQA	153	0.3856	0.6863	0.6797	0.8480	0.4058	0.6076	0.2019 [0.1372, 0.2671]	30 / 3	< 0.0001

The subset contains examples where the normalized gold-answer string is absent from the generated HyDE hypothetical passage under the rules in Table S14. All rows reuse frozen retrieval and reader answer artifacts. The diagnostic reduces the exact answer-string overlap explanation, but it does not rule out paraphrased answer content, benchmark contamination, or model memorization.

alias/paraphrase proxy, supporting-title/entity mention, or no identifiable gold content under the implemented lexical/entity rules. The second table performs a gold-aware offline retrieval-only ablation by deleting exact answer strings, and then deleting exact answer strings plus gold supporting-title strings, from the generated hypothetical passage before retrieval. These scrubbed rows are diagnostic only because gold answers and support titles are unknown at test time.

Table S18: Automatic gold-content stratification of HyDE hypothetical passages.

Dataset	Stratum	<i>N</i>	Dense all	HyDE all	Δ all [95% CI]	Dense recall	HyDE recall	Dense F1	HyDE F1	Δ F1 [95% CI]
HotpotQA	Exact-answer present	292	0.6644	0.9452	0.2808 [0.2295, 0.3322]	0.8271	0.9726	0.7976	0.9085	0.1110 [0.0756, 0.1461]
HotpotQA	Alias/paraphrase proxy present	22	0.6818	0.9545	0.2727 [0.0909, 0.4545]	0.8409	0.9773	0.7072	0.7841	0.0768 [-0.0379, 0.2078]
HotpotQA	Supporting-entity present	154	0.6104	0.7987	0.1883 [0.1234, 0.2532]	0.8019	0.8961	0.6238	0.6786	0.0548 [-0.0006, 0.1057]
HotpotQA	No identifiable gold content	32	0.3750	0.4375	0.0625 [0.0000, 0.1562]	0.6406	0.6406	0.4828	0.5693	0.0865 [-0.0312, 0.2167]
2Wiki	Exact-answer present	347	0.3458	0.6599	0.3141 [0.2680, 0.3631]	0.6650	0.8617	0.6520	0.7824	0.1304 [0.0916, 0.1738]
2Wiki	Alias/paraphrase proxy present	20	0.5500	0.9500	0.4000 [0.1500, 0.6500]	0.7750	0.9750	0.5375	0.7786	0.2411 [0.0571, 0.4500]
2Wiki	Supporting-entity present	132	0.3636	0.6515	0.2879 [0.2121, 0.3712]	0.6667	0.8314	0.3889	0.5863	0.1974 [0.1304, 0.2642]
2Wiki	No identifiable gold content	1	0.0000	0.0000	0.0000 [0.0000, 0.0000]	0.5000	0.5000	0.0000	0.0000	0.0000 [0.0000, 0.0000]

The strata are mutually exclusive and assigned in the order shown: exact normalized gold answer, automatic alias/paraphrase proxy, supporting-title/entity mention, and no identifiable gold content. The proxy is lexical and conservative; it does not replace human leakage annotation.

Table S19: Gold-aware scrubbed HyDE retrieval-only diagnostic.

Dataset	Retrieval query	All hit@10	Recall@10	Δ all vs Dense [95% CI]	Δ all vs original HyDE [95% CI]
HotpotQA	Dense question only	0.6300	0.8080	0.0000 [0.0000, 0.0000]	-0.2380 [-0.2760, -0.2000]
HotpotQA	Original HyDE q+h	0.8680	0.9280	0.2380 [0.2020, 0.2780]	0.0000 [0.0000, 0.0000]
HotpotQA	Answer-scrubbed HyDE q+h	0.8380	0.9120	0.2080 [0.1720, 0.2460]	-0.0300 [-0.0460, -0.0160]
HotpotQA	Answer+support-title-scrubbed HyDE q+h	0.7800	0.8830	0.1500 [0.1140, 0.1860]	-0.0880 [-0.1140, -0.0620]
2Wiki	Dense question only	0.3580	0.6695	0.0000 [0.0000, 0.0000]	-0.3100 [-0.3520, -0.2700]
2Wiki	Original HyDE q+h	0.6680	0.8575	0.3100 [0.2700, 0.3520]	0.0000 [0.0000, 0.0000]
2Wiki	Answer-scrubbed HyDE q+h	0.6480	0.8445	0.2900 [0.2500, 0.3340]	-0.0200 [-0.0340, -0.0060]
2Wiki	Answer+support-title-scrubbed HyDE q+h	0.5640	0.7770	0.2060 [0.1660, 0.2480]	-0.1040 [-0.1320, -0.0780]

The scrubbed rows are offline diagnostics that delete gold-aware strings from the generated hypothetical passage before retrieval. They are not deployable inference methods because gold answers and supporting titles are unavailable at test time. Reader answers are not regenerated for scrubbed rows.

Table S20: Retrieval-only HyDE query-composition check on the released IRCOT 2WikiMultihopQA test_subsampled split.

Query used for dense retrieval	<i>N</i>	Any hit@10	All hit@10	Recall@10
Question only	500	0.9840	0.3580	0.6695
Hypothetical passage only	500	0.9940	0.6640	0.8530
Question + hypothetical passage	500	0.9940	0.6680	0.8575

This no-new-LLM diagnostic reuses the existing 2Wiki HyDE generations and dense index, without additional reader calls. Complete supporting-title retrieval is higher for the hypothetical-only row than for question-only retrieval, while keeping the original question provides a small additional anchor.

Table S21: Query-token length audit and length-matched HyDE retrieval sensitivity on HotpotQA. Token counts use the MiniLM tokenizer. The hypothetical-only control matches the rewritten-query length; the question-plus-hypothetical control truncates only the hypothetical passage so that the serialized query is length-matched to question plus rewrite. No new LLM calls are made.

Query	Mean tokens	Median	P95	Hits	256-token cap
q	19.9140	19.0000	33		0
r	10.9380	10.0000	17		0
q+r	35.8520	34.0000	52		0
h	68.7400	68.0000	92		0
q+h	92.6540	92.0000	119		0

Comparator	Target retrieval query	Any-hit@10	All-hit@10	Recall@10	Δ All-hit@10	Δ Recall@10
Single rewritten query	Length-matched hypothetical only	0.9200	0.2740	0.5970	-0.3440 [-0.3940, -0.2980]	-0.2020 [-0.2310, -0.1730]
Question + rewritten query	Length-matched question + hypothetical	0.9740	0.6400	0.8070	0.0120 [-0.0240, 0.0480]	0.0040 [-0.0170, 0.0250]

Deltas are paired against the comparator shown in the first column: the hypothetical-only control is compared with the single rewritten query, and the question-plus-hypothetical control is compared with the question-plus-rewritten-query control. The question-plus-hypothetical row matches the serialized query length to the corresponding question-plus-rewrite query, up to tokenizer discreteness, while preserving the original question as an anchor in this length-matched setting. The sensitivity controls show that matched-length rewriting does not recover the full HyDE result, so the safest interpretation is a combined effect of document-like generation and a larger semantic information budget rather than document-like form alone.

S11 Query-Length Matched HyDE Sensitivity

The matched-call query-reformulation control equalizes query-side LLM invocation count but not generated-text length or information budget. The following no-new-LLM audit reports MiniLM tokenizer lengths for the main query-composition variants. For the hypothetical-only control, the HyDE hypothetical passage is truncated to the corresponding rewritten-query token count. For the question-plus-hypothetical control, only the hypothetical passage is truncated, and the serialized query is matched to the corresponding question-plus-rewrite token length. The paired deltas use the comparator shown in Table S21: the hypothetical-only row is compared with the single rewritten query, whereas the question-plus-hypothetical row is compared with the question-plus-rewritten-query control. This analysis is a query-composition sensitivity check: it does not replace the main HyDE result, but it shows that matched-length rewriting alone does not recover the full HyDE result.

Table S22: Single-annotator EM-mismatch diagnostic on HotpotQA.

Diagnostic category	Count	Share of 156 EM mismatches
Retrieval failure	6	3.8%
Partial evidence / missing hop	39	25.0%
Reader reasoning failure despite complete evidence	20	12.8%
Annotator-judged surface-form, alias, or granularity mismatch	91	58.3%

These counts describe residual exact-match mismatches in the final HyDE-style RAG + guarded verifier run. They come from a single-annotator diagnostic and are not a validated human-annotation benchmark.

S12 Error Taxonomy Rules

The annotation unit for the error taxonomy is one exact-match mismatch from the final HyDE-style RAG plus guarded verifier HotpotQA run. Each mismatch is assigned to one primary diagnostic category using frozen retrieval and answer artifacts: retrieval failure if no required supporting title is retrieved; partial evidence or missing hop if at least one but not all supporting titles are retrieved; reader reasoning failure if all supporting titles are retrieved but the annotator judged the answer semantically wrong; and surface-form, alias, or granularity mismatch if the annotator judged the retrieved evidence to support an acceptable answer that benchmark normalization still rejects by surface form or granularity.

When multiple explanations are plausible, the precedence order is retrieval availability, then reader reasoning, then format or alias. Ambiguous cases are resolved conservatively as not verifier-fixable. The annotations are used only for interpretation and case selection, not for training, prompting, or model selection. The manual inspection stage was performed by a single annotator, with no independent second-pass adjudication or inter-annotator agreement estimate, so the taxonomy should be treated as a descriptive diagnostic rather than a formally validated annotation benchmark.

Table S23: Local candidate-corpus statistics for the released processed splits.

Statistic	HotpotQA	2WikiMultihopQA
Questions	500	500
Total candidate paragraphs	4977	5000
Unique titles	4931	3273
Mean / median source-pool records contributed per question	10.0 / 10	10 / 10
Source-pool records contributed per question, min–max	3–10	10–10
Mean gold supporting titles	2.00	2.42
Mean supporting paragraphs	2.00	2.42
Mean / median distractor paragraphs	8.0 / 8	7.6 / 8
Corpus organization	Joint index over local pools	Joint index over local pools

Source-pool records are the records contributed by each released question-level candidate pool before all records are unioned into a dataset-level joint index. Retrieval is performed over the full joint corpus, not over a separate per-question pool. Exact duplicate title–paragraph records are retained before indexing; if the same paragraph appears in multiple question-level pools, each occurrence remains a separate document with a source-question-specific document identifier. Dense, BM25, and hybrid retrieval encode only document titles and paragraph texts. Question identifiers, supporting-evidence flags, gold answers, and evaluation metadata are retained in JSONL artifacts for analysis but are excluded from retrieval representations.

In both released processed splits, each annotated supporting title is associated with exactly one gold supporting paragraph record for every question, which explains why the mean supporting-title and supporting-paragraph counts are identical.

Supporting-title metrics collapse duplicate retrieved titles before computing any-hit@10, all-hit@10, and supporting-title recall@10.

Table S24: Supporting-title completeness and reader accuracy on the released IRCot HotpotQA test_subsampled split.

Scope	Supporting-title bucket	<i>N</i>	EM	F1	Mean title recall
All retrieval-only method-example observations	full title support	2181	0.7240	0.8584	1.0000
All retrieval-only method-example observations	partial title support	780	0.3846	0.4744	0.5000
All retrieval-only method-example observations	no title support	39	0.0769	0.0872	0.0000
HyDE-style RAG only	full title support	434	0.7396	0.8720	1.0000
HyDE-style RAG only	partial title support	60	0.3500	0.4472	0.5000
HyDE-style RAG only	no title support	6	0.0000	0.0000	0.0000

Full title support means all annotated supporting titles are retrieved in the top-10 evidence set; partial title support means at least one but not all supporting titles are retrieved. The pooled rows are descriptive method-example observations across the six retrieval-only methods and therefore are not treated as independent examples.

S13 Corpus, Supporting-Title Completeness, and Case Artifacts

The pooled supporting-title completeness rows are descriptive method-example observations and are not treated as statistically independent samples. The qualitative cases were selected from rule-filtered candidates using frozen per-example artifacts and are intended to make the retrieval-centered evidence changes and downstream answer-format boundaries inspectable. Table S26 gives the compact reader-facing summary, and Table S27 exposes the corresponding generated passage, retrieved titles, evidence excerpts, reader answer, verifier boundary when invoked, and guarded final answer. The supporting-paragraph audit at the end of this section rescores the same retrieval files with exact normalized title–paragraph-text keys derived from `is_supporting=true` corpus records.

Table S25: Reproducibility checklist and canonical artifact ledger.

Component	Setting reported in the paper	Audit artifact or reproducibility check
Data splits	Released IRCot processed HotpotQA and 2WikiMultiHopQA test_subsampled splits; 500 examples per split.	Upstream repository, commit, source filenames, local download timestamps, raw SHA-256 hashes, converted question/corpus JSONL SHA-256 hashes, and conversion-script hash are listed in Table S2. The processed local-corpus protocol is documented in docs/reproducibility_protocol.md.
Evidence pool	Local candidate evidence pools from the released processed data; paragraph-level documents with title plus paragraph text.	Corpus statistics are reported in the main manuscript and generated by src/evaluation/analyze_corpus_statistics.py; results are not reported as full-wiki IRCot reproduction.
Retrieval settings	Top- k = 10 final evidence documents; main dense retrieval uses all-MiniLM-L6-v2, normalized embeddings, and FAISS inner-product search; post-hoc BGE-base diagnostics use the shared configuration in Section S7.7.	Retriever scripts in src/retriever; dense, BM25, hybrid, multi-query, iterative, HyDE, BGE query-composition, and equal-budget BGE retrieval outputs are retained in local-artifacts/results and local-artifacts. Multi-query and iterative rows are heuristic controls, not faithful public-method reproductions.
Prompt settings	HyDE generation, evidence-grounded short-answer reading, and conservative verification use fixed prompt templates; reader/verifier evidence blocks are capped at 900 characters per document.	Prompt builders in src/generator and src/verifier; prompt JSONL rows store metadata for evaluation, but gold answers are not inserted into the model prompt text.
LLM settings	Hosted Qwen3.7-Max through the Alibaba Cloud Bailian/DashScope OpenAI-compatible chat-completions interface with temperature=0.0. Reported primary reader, query-generation, and verifier artifacts use the API model string qwen3.7-max. The targeted equal-budget reader robustness check uses qwen-turbo over fixed retrieved evidence. The configured max_tokens budgets are 64 for HyDE passage generation, rewritten-query generation, reader answers, and verifier JSON output; temperature 0 is not claimed to ensure exact replay.	Generation script records model, prompt, retrieved evidence, prediction, and gold-answer metadata in answer JSONL files; --resume prevents duplicate regeneration. The local invocation ledger is frozen in docs/model_invocation_ledger.md, and Table S10 reports provider/interface, hosted-alias, decoding, retry, prompt-hash, and raw-output boundaries.
Canonical ledger	Main HotpotQA table uses the audited qwen3.7-max artifact ledger for reported reader comparisons. Rule-based Evidence-Expanded Iterative Retrieval is the fixed heuristic control reported in the main table; Single-query Reformulation is the matched-call mechanism control.	Canonical ledger: local-artifacts/ircot_test500_qwen_results.csv; audit note: local-artifacts/stage1_experiment_result_audit_2026-07-06.md. Table S1 separates the primary frozen study, post-hoc robustness diagnostics, and same-split verifier tuning.
Additional analyses	Corpus statistics, HyDE query-composition diagnostic, strict equal-budget query-form diagnostic, targeted Qwen-Turbo reader check, supporting-paragraph audit, top- k sensitivity and worst-support-rank audit, corpus-scale retrieval stress test, corpus-scale leakage/hard-negative audit, answer-string-overlap diagnostic, paired reliability checks, evidence buckets, and efficiency accounting.	Stage notes and CSVs in local-artifacts; equal-budget artifacts are under local-artifacts/equal_budget_query_diagnostic; the Qwen-Turbo paired reader statistics are summarized in equal_budget_cross_reader_contrasts.csv and equal_budget_cross_reader_paired_stats.md by src/evaluation/summarize_equal_budget_reader.py; the supporting-paragraph audit is under local-artifacts/supporting_paragraph_audit and generated by src/evaluation/analyze_supporting_paragraph_metrics.py; top- k artifacts are under local-artifacts/topk_sensitivity and summarized by src/evaluation/analyze_topk_sensitivity.py; corpus-scale stress-test artifacts are under local-artifacts/corpus_scale_stress and generated by src/evaluation/analyze_corpus_scale_retrieval_stress.py; leakage-filtered and BGE-hard audit artifacts are under local-artifacts/corpus_scale_audit and generated by src/evaluation/analyze_corpus_scale_audit.py. Corpus and efficiency tables are generated from actual JSONL artifacts rather than hand-estimated counts.

The checklist is intended to make the processed-data experimental protocol auditable. Artifact freezing records the exact files used for reporting, but it is not claimed as an untouched held-out development/test split protocol for every design choice. File paths are relative to the project root when possible. API credentials are loaded from local environment variables and are not part of the reproducibility artifact.

Table S26: Artifact-grounded qualitative case studies emphasizing HyDE retrieval.

Case type	Question summary	Prediction / evidence state	Interpretation
HyDE evidence acquisition	"Ew!" is a song by a television host born where?	Dense: United States; HyDE: Bay Ridge, Brooklyn; Gold: Bay Ridge, Brooklyn; Dense support partial (1/2); HyDE support full (2/2)	HyDE retrieves the missing supporting title without an exact normalized answer-string match in the hypothetical passage, turning a partial-evidence Dense failure into a correct answer.
HyDE evidence acquisition	How many films were directed by the director of Wise Blood?	Dense: I don't know; HyDE: 37; Gold: 37; Dense support partial (1/2), missing John Huston; HyDE support full (2/2)	HyDE retrieves the director page needed for the second hop and turns a partial-support Dense abstention into a correct count answer.
Answer-standardization boundary	Which star of Zork was also the voice of Pac-Man?	HyDE: Marty Ingels; Gold: Martin Ingerman; HyDE support full (2/2)	The reader returns the professional name Marty Ingels, whereas the benchmark gold answer uses the birth name Martin Ingerman. The two names refer to the same person, illustrating that exact-match evaluation can count semantically valid aliases as errors.

Cases were selected from rule-filtered candidates using frozen HotpotQA per-example artifacts: Dense and HyDE reader outputs, supporting-title states, and HyDE answer-overlap audit rows. They are illustrative rather than additional evaluation examples.

Table S27: Compact artifact blocks for the qualitative case studies.

Example	Generated / retrieved artifacts	Evidence excerpts	Answer artifacts
HyDE evidence acquisition 5ae68fcb5542992ae0d1635b	HyDE passage: "Ew!" is a 2014 comedy song performed by American television host Jimmy Fallon, featuring will.i.am. The track is based on a recurring sketch from The Tonight Show Starring Jimmy Fallon. Jimmy Fallon, the television host and comedian who created and perform... Dense top: Ew!; Richie Kavanagh; Ezequiel Echeverría; supporting-title hits: Ew!; missing: Jimmy Fallon HyDE top: Ew!; Jimmy Fallon; Tom Ewell; supporting-title hits: Ew!; Jimmy Fallon; missing: none	[Ew!] "Ew!" is a song by American television host and comedian Jimmy Fallon, featuring American rapper will.i.am. The song is based on a sketch from "The Tonight Show Starri...; [Jimmy Fallon] James Thomas Fallon (born September 19, 1974) is an American comedian, actor, television host, and musician. He is known for his work in television as a cast member on...	Initial reader answer: Bay Ridge, Brooklyn Verifier boundary: not invoked for this example Final guarded answer: Bay Ridge, Brooklyn
HyDE evidence acquisition 5ab3f6b9554299753aec5a03	HyDE passage: The 1979 film Wise Blood was directed by acclaimed American filmmaker John Huston. Over the course of his extensive career in Hollywood, Huston directed a total of 37 feature films, including celebrated classics such as The Maltese Falcon, The Treasure of t... Dense top: Wise Blood (film); Wise Blood; Benedict Fitzgerald; supporting-title hits: Wise Blood (film); missing: John Huston HyDE top: Wise Blood (film); John Huston; Benedict Fitzgerald; supporting-title hits: John Huston; Wise Blood (film); missing: none	[Wise Blood (film)] Wise Blood is an American 1979 drama film directed by John Huston and based on the 1952 novel "Wise Blood" by Flannery O'Connor. It was filmed mostly in and around Mac...; [John Huston] John Marcellus Huston (; August 5, 1906 – August 28, 1987) was an American film director, screenwriter and actor. He wrote the screenplays for most of the 37 feature...	Dense answer: I don't know; HyDE answer: 37. Dense has partial support; HyDE has full support. Verifier boundary: raw verifier proposed "37 feature films"; guard kept 37.
Answer-standardization boundary 5a728f015542991f9a20c4e4	HyDE passage: Marty Ingels, an American actor and voice artist best known for providing the original voice of the titular character in the 1982 animated television series Pac-Man, also starred in the live-action interactive video game Return to Zork. Dense top: Marty Ingels; Mario Castañeda; David Acord; supporting-title hits: Marty Ingels; missing: Zork: Grand Inquisitor HyDE top: Marty Ingels; Milogost Reczek; Mario Castañeda; supporting-title hits: Marty Ingels; Zork: Grand Inquisitor; missing: none	[Marty Ingels] Martin Ingerman (March 9, 1936 – October 21, 2015), known professionally as Marty Ingels, was an American actor, comedian, comedy sketch writer and theatrical agent, w...; [Zork: Grand Inquisitor] Zork: Grand Inquisitor is a graphic adventure game developed by Activision and released for Windows in 1997, and for Macintosh in 2001. It builds upon the "Zork" and "...	Initial reader answer: Marty Ingels Verifier boundary: not invoked for this example Final guarded answer: Marty Ingels

Each block is a compact projection of frozen per-example artifacts. "Not invoked" means the verifier was not run for that example under the risk-selected verification protocol.

Table S28: Supporting-paragraph audit for reported title-level retrieval rows.

Dataset	Family	Method	Title all@10	Paragraph all@10	Paragraph recall@10	Gap
HotpotQA	Main	Dense	0.6300	0.6300	0.8080	+0.0000
HotpotQA	Main	Multi-query	0.6300	0.6300	0.8080	+0.0000
HotpotQA	Main	BM25	0.7320	0.7320	0.8590	+0.0000
HotpotQA	Main	Hybrid	0.7480	0.7480	0.8690	+0.0000
HotpotQA	Main	Rule-based iterative	0.7540	0.7540	0.8700	+0.0000
HotpotQA	Main	Direct rewrite	0.6180	0.6180	0.7990	+0.0000
HotpotQA	Main	Question + rewrite	0.6280	0.6280	0.8030	+0.0000
HotpotQA	Main	Hypothetical-only	0.8480	0.8480	0.9180	+0.0000
HotpotQA	Main	HyDE	0.8680	0.8680	0.9280	+0.0000
HotpotQA	Equal-budget	Keyword/entity	0.9240	0.9240	0.9570	+0.0000
HotpotQA	Equal-budget	Direct rewrite	0.8800	0.8800	0.9330	+0.0000
HotpotQA	Equal-budget	Decomposition	0.8740	0.8740	0.9320	+0.0000
HotpotQA	Equal-budget	Document-like	0.9260	0.9260	0.9600	+0.0000
2WikiMultihopQA	Main	Dense	0.3580	0.3580	0.6695	+0.0000
2WikiMultihopQA	Main	BM25	0.4320	0.4320	0.7250	+0.0000
2WikiMultihopQA	Main	Hybrid	0.4480	0.4480	0.7330	+0.0000
2WikiMultihopQA	Main	Hypothetical-only	0.6640	0.6640	0.8530	+0.0000
2WikiMultihopQA	Main	HyDE	0.6680	0.6680	0.8575	+0.0000
2WikiMultihopQA	Equal-budget	Keyword/entity	0.6760	0.6760	0.8410	+0.0000
2WikiMultihopQA	Equal-budget	Direct rewrite	0.4760	0.4760	0.7190	+0.0000
2WikiMultihopQA	Equal-budget	Decomposition	0.4540	0.4540	0.7295	+0.0000
2WikiMultihopQA	Equal-budget	Document-like	0.7520	0.7520	0.9035	+0.0000

The audited rows are exactly the rows listed in the table. They include the HotpotQA MiniLM main-result, query-composition, heuristic-control retrieval rows, the 2WikiMultihopQA reported MiniLM retrieval rows with available artifacts, and the four equal-budget query-form rows for both datasets. Title all@10 matches the main all-support hit@10 metric: all annotated supporting titles must appear in the top-10 retrieved evidence. Paragraph all@10 is stricter: all gold corpus records marked `is_supporting=true` for the question must be recovered as exact normalized title-paragraph text. The gap is title all@10 minus paragraph all@10.

In the released processed splits, each annotated supporting title is associated with one supporting paragraph record, so identical title-level and paragraph-level values are expected when the retrieved title corresponds to the same processed paragraph text. The processed artifacts do not expose gold supporting sentence text in the 500-example JSONL questions used here, so this audit cannot compute sentence-level supporting-fact coverage. It primarily verifies exact processed-record recovery rather than introducing an independent sentence-level evidence notion.

S14 Code and Artifact Availability

The anonymized review repository is available at:

```
https://github.com/creator-perterson/  
hyde-multihop-rag-reproducibility
```

It contains the implementation, prompt templates, frozen query and answer summaries, retrieval summaries, paired-analysis scripts, configuration-provenance records, model-invocation ledgers, answer-overlap diagnostics, and deduplication diagnostics, subject to the final venue policy and the licenses or terms of the upstream datasets and hosted-model provider. The release is intended to support artifact-level auditing and rescoring of the reported tables, rather than byte-identical replay of a hosted model alias.

The releasable artifact package will include local prompt text, retrieved-evidence serialization, local prediction text, parsed verifier JSON fields, stored model-string fields, artifact freeze timestamps, hashes, table-generation scripts, and aggregate analysis outputs. API credentials, private endpoint URLs, provider-side request identifiers, billing exports, latency logs, and provider raw request/response envelopes are not included because they are either private, were not retained by the local wrapper, or are not required for table-level rescoring. If venue, dataset, or hosted-provider terms restrict redistribution of generated raw text or third-party candidate-pool passages, the repository will instead provide file manifests, SHA-256 hashes, derived per-example statistics, parsed predictions/final answers where permitted, and scripts that reproduce the reported tables from authorized local copies.